

Equilibrium Operators, Variational Inequalities and Welfare:

What Is and Is Not a Theorem

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Abstract

Research line EQO001 asked whether a formal relationship between equilibrium operators, variational inequalities and welfare criteria could yield results beyond the existing ones, and its public record states that the formulation reviewed “combines known components without yet establishing an independent theorem”. This note is the definitive version of that assessment. We fix the setting (an operator F on a closed convex set K , equilibria as solutions of $\text{VI}(F, K)$, welfare as a differentiable functional W on K) and prove in full, with the classical attributions, the four facts on which any such relationship rests: (i) when F is a gradient field the equilibria are exactly the stationary points of the potential, and its minimisers when the potential is convex (Beckmann–McGuire–Winsten, Rosen, Monderer–Shapley); (ii) strict monotonicity gives uniqueness and strong monotonicity gives existence with a convergent projection iteration (Kinderlehrer–Stampacchia, Facchinei–Pang); (iii) equilibria do not in general maximise utilitarian welfare, with Pigou’s example and the bound $4/3$ for affine costs (Roughgarden–Tardos); (iv) an equilibrium is a W -maximiser only if $\nabla W(x^*)$ and $-F(x^*)$ both lie in the normal cone $N_K(x^*)$, which on the face of a simplex means that both are constant on the used coordinates, and whose repair is marginal-cost pricing. The one small addition is a characterisation of the set $\Lambda(x^*)$ of welfare weights $\lambda \geq 0$ for which a given Nash equilibrium maximises $\sum_i \lambda_i u_i$: it is always a closed convex cone contained in a first-order cone $\Lambda_1(x^*)$ defined by $F(x^*)$, the normal cone and the marginal externalities, with equality when the payoffs are jointly concave; we also show that (F, K) alone never determines $\Lambda(x^*)$. A two-player quadratic example gives the cone in closed form (a wedge whose opening depends on the product of the externalities), and a projected-gradient solver on 400 random monotone games confirms equality in every concave instance and exhibits certified strict inclusions in 56% of the non-concave instances with a non-trivial cone. No new equilibrium concept and no general equilibrium solution are offered; the value of the note is the separation of what is classical from what is open, closed by two precise open questions.

1 Introduction

The variational-inequality formulation of equilibrium is one of the unifying ideas of applied mathematics. A Wardrop traffic equilibrium [23], a Nash equilibrium of a concave game [15, 19] and a Walrasian or spatial price equilibrium can all be written as: find x^* in a closed convex set K with $\langle F(x^*), x - x^* \rangle \geq 0$ for all $x \in K$ [3, 5, 7, 9, 11, 14]. Welfare, on the other hand, is an optimisation problem: maximise a functional W over the same K . The temptation that motivated research line EQO001 is obvious: an inequality and an optimisation problem posed on the same set look like two faces of one object, and it is natural to look for a “formal relationship” from which welfare statements about equilibria would follow.

The public record of the line is candid: the formulation reviewed combined known components without establishing an independent theorem, and the line was paused. This note discharges

the line’s objective in the only way that seems honest to us: by writing down, with complete proofs where they are short, what is a theorem in this area, what is a definition in disguise, and what remains open. We add one small result on the set of welfare weights that make a given equilibrium optimal, which we believe is new only in its packaging, and we say so.

Contributions. Section 2 fixes the setting and proves the two dictionary lemmas (Nash and Wardrop equilibria as VI solutions). Section 3 proves the classical results: potentials (Theorem 3.2), monotonicity (Theorem 3.5), and the gap between equilibrium and welfare (Theorem 3.8). Section 4 characterises when an equilibrium is a welfare maximiser (Proposition 4.1) and derives marginal-cost pricing as a corollary (Corollary 4.2). Section 5 contains the small new result (Theorem 5.1), the impossibility statement that (F, K) alone does not determine welfare (Proposition 5.3), and two exactly computed examples. Section 6 reports the numerical checks with predefined criteria, Section 7 lists every claim with its status, and Section 8 states limitations and two open questions.

Scope. Everything is finite-dimensional and deterministic; no empirical data are used and no new solution concept is proposed. Every number in the text is produced by the scripts described in Appendix A, with fixed seeds.

2 Setting

Throughout, $K \subseteq \mathbb{R}^n$ is a nonempty closed convex set, $U \supseteq K$ an open set and $F: U \rightarrow \mathbb{R}^n$ a continuous map, the *equilibrium operator*. We write $\langle \cdot, \cdot \rangle$ for the Euclidean inner product.

Definition 2.1 (Variational inequality, normal cone). $\text{VI}(F, K)$ is the problem of finding $x^* \in K$ such that $\langle F(x^*), x - x^* \rangle \geq 0$ for all $x \in K$; its solution set is $\text{SOL}(F, K)$. The *normal cone* of K at $x \in K$ is $N_K(x) = \{v \in \mathbb{R}^n : \langle v, y - x \rangle \leq 0 \ \forall y \in K\}$, a closed convex cone, equal to $\{0\}$ when x is an interior point. By definition,

$$x^* \in \text{SOL}(F, K) \iff -F(x^*) \in N_K(x^*). \quad (1)$$

Definition 2.2 (Monotonicity). F is *monotone* on K if $\langle F(x) - F(y), x - y \rangle \geq 0$ for all $x, y \in K$; *strictly monotone* if the inequality is strict for $x \neq y$; μ -*strongly monotone* if $\langle F(x) - F(y), x - y \rangle \geq \mu \|x - y\|^2$ with $\mu > 0$; and L -*Lipschitz* if $\|F(x) - F(y)\| \leq L \|x - y\|$.

Definition 2.3 (Game). A *game* has N players, strategy sets $K_i \subseteq \mathbb{R}^{n_i}$ closed and convex, $K = K_1 \times \cdots \times K_N$, and payoffs $u_i: U \rightarrow \mathbb{R}$ of class C^1 on an open $U \supseteq K$, with $u_i(\cdot, x_{-i})$ concave on K_i for every x_{-i} . A *Nash equilibrium* is $x^* \in K$ with $u_i(x^*) \geq u_i(y_i, x_{-i}^*)$ for all i and $y_i \in K_i$. The *pseudo-gradient* (equilibrium operator) is $F(x) = (-\nabla_{x_1} u_1(x), \dots, -\nabla_{x_N} u_N(x))$, where ∇_{x_i} is the gradient with respect to player i ’s own variables. The *marginal externality* of player j on player $i \neq j$ at x is $\nabla_{x_j} u_i(x) \in \mathbb{R}^{n_i}$.

Lemma 2.4 (Nash equilibria are VI solutions). *In the setting of Definition 2.3, x^* is a Nash equilibrium if and only if $x^* \in \text{SOL}(F, K)$. [classical, 19; 5]*

Proof. For a concave differentiable function ϕ on a convex set C , z maximises ϕ over C iff $\langle \nabla \phi(z), y - z \rangle \leq 0$ for all $y \in C$: necessity because $t \mapsto \phi(z + t(y - z))$ is defined on $[0, 1]$ and has right derivative $\langle \nabla \phi(z), y - z \rangle$ at 0, which must be ≤ 0 ; sufficiency because $\phi(y) \leq \phi(z) + \langle \nabla \phi(z), y - z \rangle$. Apply this to $\phi = u_i(\cdot, x_{-i}^*)$ on K_i : x^* is Nash iff $\langle -F_i(x^*), y_i - x_i^* \rangle \leq 0$ for all i and $y_i \in K_i$. Summing over i gives $\langle F(x^*), y - x^* \rangle \geq 0$ for every $y \in K$; conversely, choosing $y = (y_i, x_{-i}^*)$ in the VI recovers the i -th inequality. $\square$

Definition 2.5 (Nonatomic routing). A network has edges E , commodities $k = 1, \dots, m$ with demands $d_k > 0$ and path sets P_k , $P = \bigsqcup_k P_k$. A path flow $f \in \mathbb{R}_+^P$ is feasible if $\sum_{p \in P_k} f_p = d_k$ for each k ; K is the set of feasible flows, a product of scaled simplices. Edge flows are $f_e = \sum_{p \ni e} f_p$, edge costs $c_e: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ are continuous and nondecreasing, path costs are $c_p(f) = \sum_{e \in p} c_e(f_e)$, and the equilibrium operator is $F(f) = (c_p(f))_{p \in P}$. A *Wardrop equilibrium* is a feasible f^* such that, for every k and $p, q \in P_k$ with $f_p^* > 0$, $c_p(f^*) \leq c_q(f^*)$. The *social cost* is $C(f) = \sum_p f_p c_p(f) = \sum_e f_e c_e(f_e)$, and welfare is $W = -C$.

Lemma 2.6 (Normal cone of a face of a simplex). *Let $\Delta = \{x \in \mathbb{R}_+^P : \sum_{p \in P_k} x_p = d_k, k = 1, \dots, m\}$ and $x \in \Delta$ with support $S = \{p : x_p > 0\}$. Then*

$$N_\Delta(x) = \{v : \exists \gamma \in \mathbb{R}^m \text{ with } v_p = \gamma_k \text{ for } p \in S \cap P_k, v_p \leq \gamma_k \text{ for } p \in P_k \setminus S\}.$$

Proof. For $p, q \in P_k$ with $q \in S$ and $t > 0$ small, $y = x + t(e_p - e_q) \in \Delta$, so $v \in N_\Delta(x)$ implies $v_p \leq v_q$; exchanging p, q when both are in S gives equality on $S \cap P_k$, say $v_p = \gamma_k$, and $v_p \leq \gamma_k$ off S . Conversely, if v has this form and $y \in \Delta$, then $\langle v, y - x \rangle = \sum_k (\sum_{p \in P_k} v_p y_p - \gamma_k d_k) \leq \sum_k (\gamma_k \sum_{p \in P_k} y_p - \gamma_k d_k) = 0$, using $v_p \leq \gamma_k$ on P_k and $y \geq 0$. $\square$

Lemma 2.7 (Wardrop equilibria are VI solutions). *f^* is a Wardrop equilibrium iff $f^* \in \text{SOL}(F, K)$ with $F = (c_p)_p$. [classical, 21; 3]*

Proof. By (1) and Lemma 2.6, $f^* \in \text{SOL}(F, K)$ iff for each k there is γ_k with $c_p(f^*) = -\gamma_k$ on used paths of P_k and $c_p(f^*) \geq -\gamma_k$ on unused ones, which is Wardrop's condition. $\square$

Definition 2.8 (Welfare functionals). A *welfare functional* is a C^1 function $W: U \rightarrow \mathbb{R}$ to be maximised over K . For a game, the *weighted welfare* with weights $\lambda \in \mathbb{R}_+^N$ is $W_\lambda = \sum_i \lambda_i u_i$; $\lambda = \mathbf{1}$ is the utilitarian criterion. For routing, $W = -C$.

3 The classical results, with proofs

3.1 Potentials

Proposition 3.1 (Poincaré lemma on a convex set). *Let U be open and convex and $F: U \rightarrow \mathbb{R}^n$ of class C^1 with symmetric Jacobian $DF(x)$ at every x . Fix $x_0 \in U$ and set $P(x) = \int_0^1 \langle F(x_0 + t(x - x_0)), x - x_0 \rangle dt$. Then $P \in C^2(U)$ and $\nabla P = F$. [classical; proved here]*

Proof. Write $\gamma_t = x_0 + t(x - x_0)$ and $h = x - x_0$. Differentiating under the integral sign (the integrand is C^1 in x on the compact segment),

$$\begin{aligned} \partial_i P(x) &= \int_0^1 \left[F_i(\gamma_t) + t \sum_j \partial_i F_j(\gamma_t) h_j \right] dt = \int_0^1 \left[F_i(\gamma_t) + t \sum_j \partial_j F_i(\gamma_t) h_j \right] dt \\ &= \int_0^1 \frac{d}{dt} [t F_i(\gamma_t)] dt = F_i(x), \end{aligned}$$

where the second equality uses $\partial_i F_j = \partial_j F_i$ and the third the chain rule $\frac{d}{dt} F_i(\gamma_t) = \sum_j \partial_j F_i(\gamma_t) h_j$. $\square$

Theorem 3.2 (Equilibria of gradient operators). *Suppose $F = \nabla P$ on $U \supseteq K$ with $P \in C^1(U)$. Then*

$$\text{SOL}(F, K) = \{x \in K : \langle \nabla P(x), y - x \rangle \geq 0 \forall y \in K\},$$

the set of stationary points of P over K ; it contains $\arg \min_K P$, and equals it when P is convex on K . If P is strictly convex, $\text{SOL}(F, K)$ has at most one point. [classical: 1 for routing, 19 and 13 for games; proved here]

Proof. The first identity is Definition 2.1 with $F = \nabla P$. If x minimises P over K then, as in the proof of Lemma 2.4, the right derivative of $t \mapsto P(x + t(y - x))$ at 0 is nonnegative, which is the stationarity inequality. If P is convex and x is stationary, $P(y) \geq P(x) + \langle \nabla P(x), y - x \rangle \geq P(x)$ for all $y \in K$. If P is strictly convex its minimiser over the convex set K is unique. $\square$

Remark 3.3 (The three classical instances). (a) *Routing* [1]. With Δ the edge–path incidence matrix, $F(f) = \Delta^\top c(\Delta f)$ has Jacobian $\Delta^\top \text{diag}(c'_e(f_e))\Delta$, symmetric, and Proposition 3.1 produces $P(f) = \sum_e \int_0^{f_e} c_e(s) ds$, convex when the c_e are nondecreasing. Wardrop equilibria are the minimisers of P , and the edge flows are unique when the c_e are strictly increasing. Note that $P \neq C$: the potential integrates the cost, the social cost multiplies it; this is the whole of Section 3.3. (b) *Concave games* [19]. Lemma 2.4 plus Theorem 3.5(a) below give uniqueness under Rosen’s diagonal strict concavity, which for unit weights is strict monotonicity of the pseudo-gradient; existence on compact K is the Hartman–Stampacchia theorem quoted after Theorem 3.5. (c) *Potential games* [13]. A C^1 game is an exact potential game with potential Π iff $\nabla_{x_i} u_i = \nabla_{x_i} \Pi$ for all i , i.e. $F = -\nabla \Pi$; by Theorem 3.2 its equilibria are the stationary points of $-\Pi$ over K , and its maximisers when Π is concave. The potential is a bookkeeping device, not a welfare functional: in general $\Pi \neq \sum_i u_i$ (Example 5.5 below is a potential game whose equilibrium maximises no weighted welfare).

3.2 Monotonicity: uniqueness, existence and a solver

Lemma 3.4 (Projection). *Let Π_K be the Euclidean projection onto K . Then $z = \Pi_K(y)$ iff $z \in K$ and $\langle y - z, x - z \rangle \leq 0$ for all $x \in K$; Π_K is nonexpansive; and for every $\gamma > 0$, $x \in \text{SOL}(F, K)$ iff $x = \Pi_K(x - \gamma F(x))$. [classical, 11; proved here]*

Proof. z minimises the strictly convex $\frac{1}{2}\|y - \cdot\|^2$ over K iff (Theorem 3.2 with $P = \frac{1}{2}\|y - \cdot\|^2$) $\langle z - y, x - z \rangle \geq 0$ for all $x \in K$. For nonexpansiveness, add the characterisations at $z = \Pi_K y$, $x = \Pi_K y'$ and at $z' = \Pi_K y'$, $x = \Pi_K y$: $\langle (y - y') - (z - z'), z' - z \rangle \leq 0$, so $\|z - z'\|^2 \leq \langle y - y', z - z' \rangle \leq \|y - y'\| \|z - z'\|$. Finally $x = \Pi_K(x - \gamma F(x))$ iff $\langle -\gamma F(x), y - x \rangle \leq 0$ for all $y \in K$, which is the VI. $\square$

Theorem 3.5 (Monotone operators). (a) *If F is strictly monotone on K , $\text{SOL}(F, K)$ has at most one point.* (b) *If F is μ -strongly monotone and L -Lipschitz on K , $\text{SOL}(F, K)$ is a singleton $\{x^*\}$, and for $0 < \gamma < 2\mu/L^2$ the iteration $x_{t+1} = \Pi_K(x_t - \gamma F(x_t))$ satisfies $\|x_{t+1} - x^*\| \leq q\|x_t - x^*\|$ with $q = \sqrt{1 - 2\gamma\mu + \gamma^2 L^2} < 1$; for $\gamma = \mu/L^2$, $q = \sqrt{1 - \mu^2/L^2}$. [classical, 5, 11; proved here]*

Proof. (a) If $x, y \in \text{SOL}(F, K)$ then $\langle F(x), y - x \rangle \geq 0$ and $\langle F(y), x - y \rangle \geq 0$; adding, $\langle F(x) - F(y), x - y \rangle \leq 0$, so $x = y$ by strict monotonicity. (b) Let $T(x) = \Pi_K(x - \gamma F(x))$. By nonexpansiveness, $\|Tx - Ty\|^2 \leq \|(x - y) - \gamma(F(x) - F(y))\|^2 = \|x - y\|^2 - 2\gamma\langle F(x) - F(y), x - y \rangle + \gamma^2\|F(x) - F(y)\|^2 \leq q^2\|x - y\|^2$, and $q < 1$ iff $\gamma < 2\mu/L^2$. Banach’s fixed-point theorem on the complete set K gives a unique fixed point, which is the unique VI solution by Lemma 3.4, and the contraction estimate is the stated rate. $\square$

Remark 3.6. Existence for merely continuous F on compact convex K is the Hartman–Stampacchia theorem [9], a consequence of Brouwer’s theorem applied to T ; we do not reproduce it. All solvers used in Section 6 are the iteration of Theorem 3.5(b) with $\gamma = \mu/L^2$, so every computed equilibrium comes with a proof of uniqueness and a residual certificate.

3.3 Equilibrium is not welfare: Pigou and the price of anarchy

Example 3.7 (Pigou). Two parallel links, unit demand, $c_1 \equiv 1$ and $c_2(x) = x^d$, $d \geq 1$ [17]. Every user on link 2 pays $1 \leq c_1$, so $f^* = (0, 1)$ is the Wardrop equilibrium (unique edge flow, Remark 3.3a) with $C(f^*) = 1$. The social cost of $(1 - x, x)$ is $1 - x + x^{d+1}$, minimised at $x = (d + 1)^{-1/d}$ with value $1 - \frac{d}{d+1}(d + 1)^{-1/d}$. For $d = 1$ the optimum is $(\frac{1}{2}, \frac{1}{2})$ with cost $\frac{3}{4}$ and

ratio $\frac{4}{3}$; the ratio tends to infinity with d , since $(d+1)^{-1/d} \rightarrow 1$. The computed values are in Table 1.

Theorem 3.8 (Price of anarchy for affine costs). *If every $c_e(x) = a_e x + b_e$ with $a_e, b_e \geq 0$, then for every Wardrop equilibrium f and every feasible f' , $C(f) \leq \frac{4}{3}C(f')$. The constant is attained in Example 3.7 with $d = 1$. [classical, 20; the proof below follows 2]*

Proof. By Lemma 2.7, $\sum_p c_p(f)(f'_p - f_p) \geq 0$; regrouping by edges, $C(f) = \sum_e c_e(f_e)f_e \leq \sum_e c_e(f_e)f'_e$. For each edge, $c_e(f_e)f'_e = c_e(f'_e)f'_e + a_e(f_e - f'_e)f'_e \leq c_e(f'_e)f'_e + \frac{1}{4}a_e f_e^2 \leq c_e(f'_e)f'_e + \frac{1}{4}c_e(f_e)f_e$, since $y \mapsto a_e(f_e - y)y$ is maximised at $y = f_e/2$ and $a_e f_e^2 \leq (a_e f_e + b_e)f_e$. Summing, $C(f) \leq C(f') + \frac{1}{4}C(f)$. $\square$

4 When an equilibrium is a welfare maximiser

Proposition 4.1 (Necessary and sufficient first-order conditions). *Let $x^* \in \text{SOL}(F, K)$ and let W be differentiable at x^* .*

- (a) *If $x^* \in \arg \max_K W$ then $\nabla W(x^*) \in N_K(x^*)$ and $-F(x^*) \in N_K(x^*)$. If W is concave on K , the converse holds: $x^* \in \arg \max_K W$ iff $\nabla W(x^*) \in N_K(x^*)$.*
- (b) *If x^* is an interior point of K , (a) forces $F(x^*) = 0 = \nabla W(x^*)$. If $N_K(x^*) = \mathbb{R}_+ \nu$ is a ray and $F(x^*) \neq 0$, (a) forces $\nabla W(x^*) = -\theta F(x^*)$ for some $\theta \geq 0$.*
- (c) (Standard form.) *Let $K = \{x \in \mathbb{R}_+^n : Ax = b\}$ with $A \in \mathbb{R}^{m \times n}$ and let S be the support of x^* . If x^* is an equilibrium and a W -maximiser, there are $y, z \in \mathbb{R}^m$ with*

$$\begin{aligned} F_S(x^*) &= (A^\top y)_S, & F_j(x^*) &\geq (A^\top y)_j \quad (j \notin S), \\ \nabla_S W(x^*) &= (A^\top z)_S, & \partial_j W(x^*) &\leq (A^\top z)_j \quad (j \notin S). \end{aligned}$$

In particular, on a single simplex ($A = \mathbf{1}^\top$) both $F(x^)$ and $\nabla W(x^*)$ are constant on the support: they are proportional on the equilibrium face, and each satisfies Wardrop's principle with respect to its own coordinates.*

- (d) *If F is strictly monotone, W is strictly concave and both $\text{SOL}(F, K)$ and $\arg \max_K W$ are nonempty, then $\text{SOL}(F, K) = \arg \max_K W$ iff the unique equilibrium satisfies $\nabla W(x^*) \in N_K(x^*)$; otherwise the two sets are disjoint.*

[proved here; elementary]

Proof. (a) The first-order necessary condition for a maximum of a differentiable function over a convex set (proof of Lemma 2.4) is $\langle \nabla W(x^*), y - x^* \rangle \leq 0$ for all $y \in K$, i.e. $\nabla W(x^*) \in N_K(x^*)$; the second membership is (1); sufficiency under concavity is the same argument as in Theorem 3.2. (b) For interior x^* , $N_K(x^*) = \{0\}$. If $N_K(x^*) = \mathbb{R}_+ \nu$, write $-F(x^*) = \alpha \nu$ and $\nabla W(x^*) = \beta \nu$ with $\alpha, \beta \geq 0$; $F(x^*) \neq 0$ gives $\alpha > 0$ and $\theta = \beta/\alpha$. (c) The tangent cone of K at x^* is $\{h : Ah = 0, h_j \geq 0 \ (j \notin S)\}$ and its polar, by Farkas' lemma [18], is $N_K(x^*) = \{A^\top w - v : w \in \mathbb{R}^m, v \geq 0, v_S = 0\}$. Apply this to $-F(x^*)$ (with $y = -w$) and to $\nabla W(x^*)$ (with $z = w$). (d) Both sets are singletons by Theorem 3.5(a) and strict concavity; apply (a). $\square$

Proposition 4.1 is a statement about first-order conditions and nothing more: it says that an equilibrium can be a welfare optimum only if the two problems happen to share their KKT point, and it gives no mechanism producing that coincidence. The one mechanism that always works is to change the operator.

Corollary 4.2 (Marginal-cost pricing and the toll set). *Let $C \in C^1(U)$ be a cost to be minimised over K and define the toll field $\tau = \nabla C - F$. Then*

$$\text{SOL}(F + \tau, K) = \{x \in K : -\nabla C(x) \in N_K(x)\} \supseteq \arg \min_K C,$$

with equality when C is convex on K , and a singleton when C is strictly convex; the tolled operator is a gradient field with potential C . For separable routing costs, $\tau_p(f) = \sum_{e \in p} f_e c'_e(f_e)$, the Pigou–Knight–Beckmann marginal-cost toll. More generally, for a target $\bar{x} \in K$ the set of constant toll vectors $t \in \mathbb{R}^n$ with $\bar{x} \in \text{SOL}(F + t, K)$ is the translated cone $-F(\bar{x}) - N_K(\bar{x})$, polyhedral when K is. [classical, 1, 10, 12, 17; proved here]

Proof. $F + \tau = \nabla C$, so the first statement is Theorem 3.2 with $P = C$. For routing, $\partial C / \partial f_p = \sum_{e \in p} (c_e(f_e) + f_e c'_e(f_e))$, whence τ_p . Finally $\bar{x} \in \text{SOL}(F + t, K)$ iff $-F(\bar{x}) - t \in N_K(\bar{x})$ by (1). $\square$

5 The cone of welfare weights

Fix a game as in Definition 2.3 and a Nash equilibrium x^* . Write $u = (u_1, \dots, u_N) : K \rightarrow \mathbb{R}^N$ and $G(x^*) = Du(x^*) \in \mathbb{R}^{N \times n}$ for the Jacobian of the payoff vector, whose i -th row is $\nabla u_i(x^*)^\top$ (all variables, own and others'). Define

$$\Lambda(x^*) = \{\lambda \in \mathbb{R}_+^N : x^* \in \arg \max_K W_\lambda\}, \quad \Lambda_1(x^*) = \{\lambda \in \mathbb{R}_+^N : G(x^*)^\top \lambda \in N_K(x^*)\}.$$

For a set $D \subseteq \mathbb{R}^N$, $D^\circ = \{\lambda : \langle \lambda, d \rangle \leq 0 \ \forall d \in D\}$ is its polar cone.

Theorem 5.1 (Welfare-weight cone). (a) $\Lambda(x^*) = \mathbb{R}_+^N \cap \{u(x) - u(x^*) : x \in K\}^\circ$. In particular $\Lambda(x^*)$ is a closed convex cone.

(b) $\Lambda(x^*) \subseteq \Lambda_1(x^*)$, and $\Lambda_1(x^*)$ is a closed convex cone, polyhedral whenever K is polyhedral.

(c) If every u_i is concave on K (jointly in all variables), then $\Lambda(x^*) = \Lambda_1(x^*)$. More generally, equality holds whenever W_λ is concave on K for every $\lambda \in \Lambda_1(x^*)$.

(d) (Block form.) $\lambda \in \Lambda_1(x^*)$ iff for every player j

$$-\lambda_j F_j(x^*) + \sum_{i \neq j} \lambda_i \nabla_{x_j} u_i(x^*) \in N_{K_j}(x_j^*). \quad (2)$$

Since $-F_j(x^*) \in N_{K_j}(x_j^*)$ already, a sufficient condition is that the λ -weighted externality $\sum_{i \neq j} \lambda_i \nabla_{x_j} u_i(x^*)$ lies in $N_{K_j}(x_j^*)$ for every j ; when x_j^* is interior to K_j , (2) says exactly that this weighted externality vanishes.

(e) $\Lambda(x^*) \neq \{0\}$ implies that x^* is weakly Pareto efficient in K (no $x \in K$ with $u_i(x) > u_i(x^*)$ for all i); under the concavity of (c), the converse holds.

[proved here; (a) and (e) are the standard scalarisation argument of multi-objective optimisation [4, 6], specialised to Nash equilibria]

Proof. (a) $\lambda \in \Lambda(x^*)$ iff $\lambda \geq 0$ and $\langle \lambda, u(x) - u(x^*) \rangle \leq 0$ for all $x \in K$, which is the stated intersection; polar cones and the orthant are closed convex cones. (b) If x^* maximises the differentiable W_λ over K , the first-order condition gives $\nabla W_\lambda(x^*) = \sum_i \lambda_i \nabla u_i(x^*) = G(x^*)^\top \lambda \in N_K(x^*)$. $\Lambda_1(x^*)$ is the preimage of the closed convex cone $N_K(x^*)$ under a linear map, intersected with $\mathbb{R}_+^N$; when K is polyhedral so is $N_K(x^*)$, hence so is $\Lambda_1(x^*)$. (c) For $\lambda \geq 0$, W_λ is concave when the u_i are, and the first-order condition is then sufficient (Proposition 4.1(a)); the general statement is the same argument. (d) For a product set, $v \in N_K(x^*)$ iff $v_j \in N_{K_j}(x_j^*)$ for each j : taking $y \in K$ that differs from x^* in block j only shows necessity, and summing the

block inequalities shows sufficiency. Block j of $G(x^*)^\top \lambda$ is $\sum_i \lambda_i \nabla_{x_j} u_i(x^*) = \lambda_j \nabla_{x_j} u_j(x^*) + \sum_{i \neq j} \lambda_i \nabla_{x_j} u_i(x^*)$ and $\nabla_{x_j} u_j(x^*) = -F_j(x^*)$. The sufficient condition follows because $N_{K_j}(x_j^*)$ is a convex cone containing $-F_j(x^*)$ by Lemma 2.4; for interior x_j^* , $N_{K_j}(x_j^*) = \{0\}$ forces $F_j(x^*) = 0$ and then (2) reads $\sum_{i \neq j} \lambda_i \nabla_{x_j} u_i(x^*) = 0$. (e) If $\lambda \in \Lambda(x^*) \setminus \{0\}$ and $u(x) > u(x^*)$ componentwise, then $\langle \lambda, u(x) \rangle > \langle \lambda, u(x^*) \rangle$, a contradiction. Conversely, under concavity the set $A = u(K) - \mathbb{R}_+^N$ is convex; weak efficiency means $u(x^*) + \varepsilon \mathbf{1} \notin A$ for every $\varepsilon > 0$, so $u(x^*)$ is a boundary point of A and a supporting hyperplane gives $\lambda \neq 0$ with $\langle \lambda, a \rangle \leq \langle \lambda, u(x^*) \rangle$ for all $a \in A$; since $u(x^*) - \mathbb{R}_+^N \subseteq A$, $\lambda \geq 0$, and $a = u(x)$ gives $\lambda \in \Lambda(x^*)$. $\square$

Remark 5.2. The content of Theorem 5.1 is not the scalarisation, which is textbook, but the reading of (2): the cone is cut out by the equilibrium face (through $N_{K_j}(x_j^*)$ and $F_j(x^*)$, both determined by (F, K)) and by the marginal externalities $\nabla_{x_j} u_i(x^*)$, which are *not* determined by (F, K) . The next proposition makes the second half precise.

Proposition 5.3 ((F, K) does not determine welfare). *Let the u_i be continuous on $K = \prod K_i$ with K_i compact, and let x^* be a Nash equilibrium. Define $v_i(x_{-i}) = \max_{y_i \in K_i} u_i(y_i, x_{-i})$ and $u'_i = u_i - v_i$. Then the game (u'_i) has the same best-response correspondences and the same Nash equilibria as (u_i) ; wherever $\nabla_{x_i} u_i$ exists, $\nabla_{x_i} u'_i = \nabla_{x_i} u_i$, so the two games have the same equilibrium operator; and x^* maximises every $W'_\lambda = \sum_i \lambda_i u'_i$, $\lambda \in \mathbb{R}_+^N$, over K : $\Lambda'(x^*) = \mathbb{R}_+^N$. Consequently welfare optimality of an equilibrium is not a property of (F, K) : every pair (F, K) that arises from a game is compatible with $\Lambda(x^*) = \mathbb{R}_+^N$, and the pair of Example 5.5(i) is also compatible with $\Lambda(x^*) = \{0\}$. [proved here; the function $\sum_i (v_i - u_i)$ is the Nikaidô–Isoda gap function [16]]*

Proof. v_i does not depend on x_i , so subtracting it changes neither $\arg \max_{y_i} u_i(y_i, x_{-i})$ nor $\nabla_{x_i} u_i$. By construction $u'_i \leq 0$ on K , and $u'_i(x^*) = 0$ because x_i^* is a best response to x_{-i}^* ; hence $W'_\lambda \leq 0 = W'_\lambda(x^*)$ for every $\lambda \geq 0$. The last sentence combines this with Example 5.5(i), where $\Lambda(x^*) = \{0\}$. $\square$

Example 5.4 (Two players, exact cone). Let $K = [0, 1]^2$, $\alpha > 1$, $\beta_1, \beta_2 \geq 0$, and

$$u_1(x) = \alpha x_1 - \frac{1}{2} x_1^2 - \beta_1 x_2, \quad u_2(x) = \alpha x_2 - \frac{1}{2} x_2^2 - \beta_2 x_1.$$

Each player's action harms the other linearly. The equilibrium operator is $F(x) = (x_1 - \alpha, x_2 - \alpha)$, with Jacobian I , so F is 1-strongly monotone and 1-Lipschitz and does not depend on β ; the unique equilibrium is $x^* = (1, 1)$, at which $N_K(x^*) = \mathbb{R}_+^2$ and $F(x^*) = (1 - \alpha)\mathbf{1}$. Here $G(x^*) = \begin{pmatrix} \alpha-1 & -\beta_1 \\ -\beta_2 & \alpha-1 \end{pmatrix}$, so by (2)

$$\Lambda_1(x^*) = \{\lambda \geq 0 : (\alpha - 1)\lambda_1 \geq \beta_2 \lambda_2, (\alpha - 1)\lambda_2 \geq \beta_1 \lambda_1\}.$$

Both payoffs are jointly concave (Hessians $-\text{diag}(1, 0)$ and $-\text{diag}(0, 1)$), so $\Lambda(x^*) = \Lambda_1(x^*)$ by Theorem 5.1(c). Direct check: W_λ is separable, $x_1 = 1$ maximises $\lambda_1(\alpha x_1 - \frac{1}{2} x_1^2) - \lambda_2 \beta_2 x_1$ on $[0, 1]$ iff its derivative at 1, $\lambda_1(\alpha - 1) - \lambda_2 \beta_2$, is ≥ 0 , and symmetrically. Thus the cone is a wedge, non-trivial iff $\beta_1 \beta_2 \leq (\alpha - 1)^2$ and two-dimensional iff the inequality is strict; for $\alpha = 2$, $\beta_1 = \beta_2 = \frac{1}{2}$ it is $\{\frac{1}{2} \lambda_2 \leq \lambda_1 \leq 2 \lambda_2\}$ (Figure 1): the corner equilibrium is welfare-optimal exactly when neither player is weighted more than twice the other. Two games with β and β' have the same F and different cones, which is Proposition 5.3 in miniature.

Example 5.5 (Cournot duopoly). Let $u_i(q) = q_i(a - q_1 - q_2)$ on $K = [0, 1]^2$; $\partial^2 u_1 / \partial q_1^2 = -2 < 0$, so Lemma 2.4 applies, and $F(q) = (2q_1 + q_2 - a, q_1 + 2q_2 - a)$ has symmetric part $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \succ 0$. (i) For $a = 1$ the equilibrium $q^* = (\frac{1}{3}, \frac{1}{3})$ is interior, $F(q^*) = 0$, and $\partial u_1 / \partial q_2 = -q_1^* \neq 0$: by Theorem 5.1(d) the interior condition fails for every $\lambda \neq 0$, so $\Lambda(q^*) = \{0\}$. Indeed the collusive $(\frac{1}{4}, \frac{1}{4})$ gives each firm $\frac{1}{8} > \frac{1}{9}$. Note that this is an exact potential game with $\Pi(q) = a(q_1 + q_2) - q_1^2 - q_2^2 - q_1 q_2$, whose maximiser is the equilibrium: the potential is not

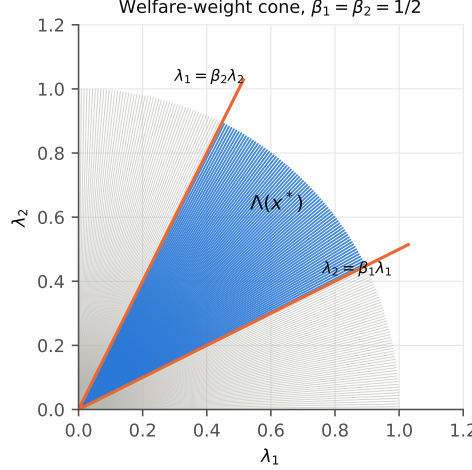


Figure 1: Example 5.4 with $\alpha = 2$, $\beta_1 = \beta_2 = 0.5$. Each ray is one of the 181 tested directions $\lambda = (\cos \theta, \sin \theta)$; blue rays are those for which the exact box-constrained maximisation of W_λ returns $x^* = (1, 1)$, grey rays those for which a better point exists. The orange lines are the closed-form boundaries $\lambda_2 = \beta_1 \lambda_1$ and $\lambda_1 = \beta_2 \lambda_2$ (at 26.57° and 63.43°). Closed form, first-order cone and exact maximisation agree on 181/181 directions.

a welfare functional. (ii) For $a = 4$ the equilibrium is the corner $(1, 1)$, $F(q^*) = (-1, -1)$, $G(q^*) = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ and $\Lambda_1(q^*) = \{\lambda_1 = \lambda_2 \geq 0\}$, a ray. The payoffs are not jointly concave, so Theorem 5.1(c) is silent; but $W_{(1,1)}(q) = (q_1 + q_2)(4 - q_1 - q_2)$ is concave, so $\Lambda(q^*) = \Lambda_1(q^*)$ nevertheless. Concavity is sufficient, not necessary.

6 Numerical checks

All computations use the projected-gradient iteration of Theorem 3.5(b) with $\gamma = \mu/L^2$, run to a fixed-point step below 10^{-12} , and report the natural residual $\|x - \Pi_K(x - F(x))\|$, which vanishes exactly at solutions (Lemma 3.4). Criteria were fixed before the runs and are stated with each experiment. The experiments verify theorems and produce descriptive statistics of a sampling distribution; they discover nothing that is not proved above, and the frequencies reported depend on the distribution of random instances.

6.1 E1: routing (Pigou, affine price of anarchy, tolls)

Protocol. E1a evaluates Example 3.7 in closed form and by a dense grid search over the split (2×10^5 points). E1b draws 200 parallel-link networks with 2–5 links, $a_e \sim U(0.05, 1)$ and b_e equal to 0 with probability $\frac{1}{2}$ and otherwise $U(0, 1)$, unit demand; equilibrium ($F = c$, $\mu = \min a_e$, $L = \max a_e$) and optimum ($F = \nabla C = 2a \circ f + b$) are computed by the solver on the simplex and, independently, by exact water-filling on the common cost level. E1c solves the tolled operator $c + \tau$ with $\tau_e = a_e f_e$ (Corollary 4.2). *Criteria.* Residuals $\leq 10^{-8}$; solver and water-filling agree to 10^{-7} ; PoA $\leq 4/3 + 10^{-9}$ in every instance and exactly $4/3$ for Pigou with $d = 1$; tolled equilibrium equals the optimum to 10^{-7} .

Results. All criteria hold. Pigou: PoA = 1.333333 for $d = 1$, 1.6258 for $d = 2$ and 4.73 for $d = 16$; closed form and grid search agree to 1.5×10^{-11} . Affine instances: maximum PoA 1.2094, mean 1.0305, median 1.0027, 0 violations of the bound; 41% of the instances have PoA > 1.01 and 12% have PoA > 1.10 ; maximum residual 3.4×10^{-11} , maximum deviation from water-filling 8.8×10^{-11} , at most 2649 iterations; the tolled equilibrium coincides with the optimum to 2.2×10^{-16} in every instance (Figure 2, Table 1).

d	optimal flow on link 2	optimal cost	price of anarchy
1	0.5000	0.7500	1.3333
2	0.5774	0.6151	1.6258
4	0.6687	0.4650	2.1505
8	0.7598	0.3246	3.0808
16	0.8377	0.2116	4.7268

Table 1: E1a. Pigou’s example with $c_1 \equiv 1$, $c_2(x) = x^d$: the equilibrium always has cost 1; the optimum splits the flow.

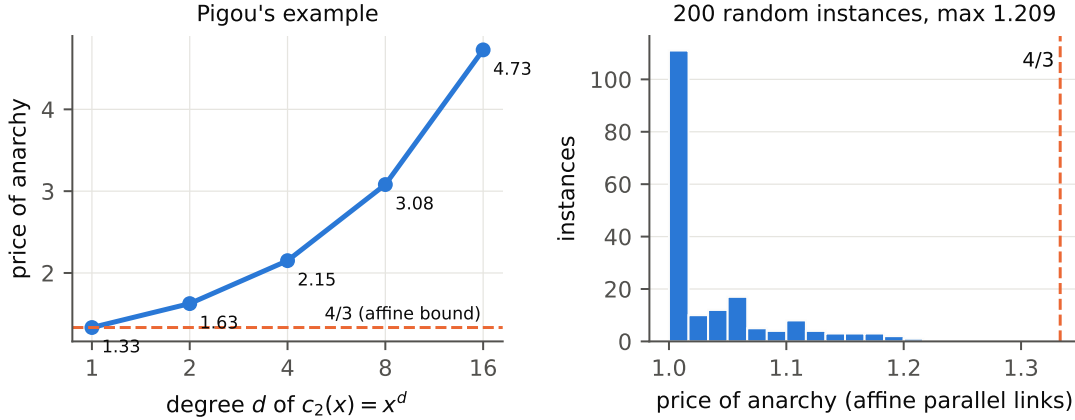


Figure 2: E1. Left: price of anarchy of Pigou’s example against the degree d (Table 1); the dashed line is the affine bound $4/3$ of Theorem 3.8. Right: price of anarchy of 200 random affine parallel-link instances; none exceeds $4/3$.

6.2 E2–E4: welfare-weight cones

Games. N players with scalar actions in $[0, 1]$ and quadratic payoffs $u_i(x) = -\frac{1}{2}x^\top H_i x + h_i^\top x$, so that $F(x) = Mx - m$ with $M_{ij} = (H_i)_{ij}$. In the *concave* family $H_i = B_i B_i^\top$ with Gaussian B_i (so each u_i is jointly concave); in the *non-concave* family H_i is a Gaussian symmetric matrix. In both, the diagonal entries $(H_i)_{ii}$ are raised by a common amount so that the symmetric part of M has least eigenvalue $\frac{1}{2}$: F is then strongly monotone, the equilibrium is unique, and $(H_i)_{ii} > 0$ ensures own-concavity, so Lemma 2.4 applies. The entries of h_i are $N(1, 1.5^2)$, which puts between 64% and 75% of the equilibrium coordinates on the boundary, depending on the family.

Exact membership tests. $\lambda \in \Lambda_1(x^*)$ is a finite system of linear equalities and inequalities (Theorem 5.1(d) with $N_{[0,1]}(0) = \mathbb{R}_-$, $N_{[0,1]}(1) = \mathbb{R}_+$, $N_{[0,1]}(t) = \{0\}$ inside), tested to 10^{-9} . $\lambda \in \Lambda(x^*)$ is decided by computing $\max_K W_\lambda$ exactly: the maximiser of a quadratic over a box lies in the relative interior of one of the 3^N faces and is stationary for the restriction to that face, so enumerating the faces and solving each stationarity system yields the maximum; $\lambda \in \Lambda(x^*)$ iff $W_\lambda(x^*) \geq \max_K W_\lambda - 10^{-8}$. A point x with $W_\lambda(x) > W_\lambda(x^*) + 10^{-6}$ is a *certified witness* that $\lambda \notin \Lambda(x^*)$. On 13 concave instances the face enumeration was cross-checked against multistart projected gradient ascent (maximum discrepancy 1.4×10^{-17}).

Geometry of Λ_1 . Non-triviality and dimension are computed by linear programming (maximising $\mathbf{1}^\top \lambda$ on the cone truncated by $\mathbf{1}^\top \lambda \leq 1$; then detecting implicit equalities row by row), and up to 12 vertices of $\Lambda_1 \cap \{\mathbf{1}^\top \lambda = 1\}$ plus their barycentre are the *cone points* tested for membership in Λ . In addition every point of a simplex grid (step $1/12$ for $N = 3$, $1/8$ for $N = 4$) is classified by both tests.

Criteria. (E2) In Example 5.4, closed form, linear test and exact maximisation must agree

N	conc.	games	$\Lambda_1 \neq \{0\}$	dim Λ_1 : games	cone pts $\in \Lambda$	grid $\in \Lambda_1$: $\in \Lambda$	grid $\notin \Lambda_1$: $\notin \Lambda$	witn.
3	yes	100	28	1:4, 2:5, 3:19	115/115	575/575	8525/8525	0
4	yes	100	33	1:2, 2:7, 3:15, 4:9	153/153	307/307	16193/16193	0
3	no	100	57	1:13, 2:14, 3:30	162/206	1025/1063	8037/8037	23
4	no	100	46	1:2, 2:2, 3:13, 4:29	152/246	1265/1337	15163/15163	35

Table 2: E3–E4. Random monotone quadratic games on $[0, 1]^N$. “conc.” is joint concavity; “cone pts” are vertices and barycentre of $\Lambda_1 \cap \{\mathbf{1}^\top \lambda = 1\}$; “witn.” counts games with a certified $\lambda \in \Lambda_1 \setminus \Lambda$. Concave games realise $\Lambda = \Lambda_1$ in every test; non-concave games realise the strict inclusion in a sizeable fraction.

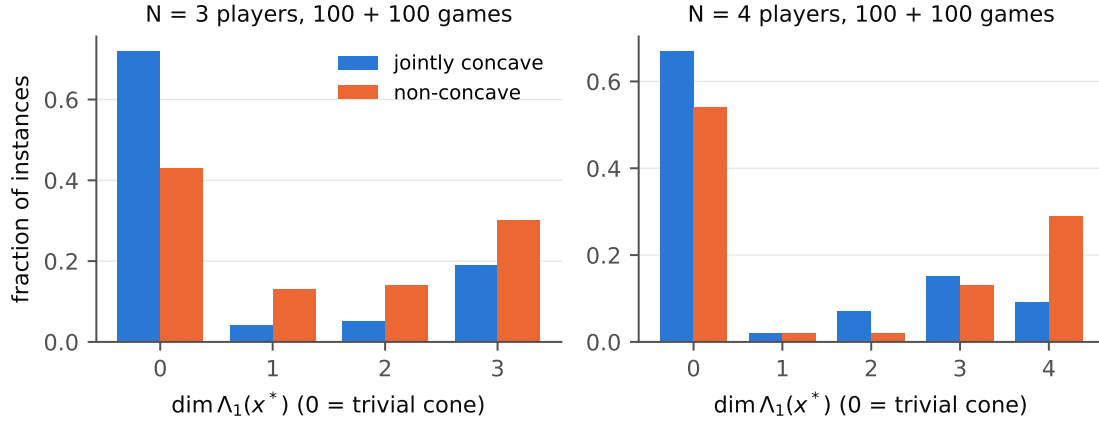


Figure 3: E3–E4. Dimension of the first-order cone $\Lambda_1(x^*)$ across random games (0 means $\Lambda_1 = \{0\}$: the equilibrium is welfare-optimal for no weights). The dimension is determined by the equilibrium face and the externalities (Theorem 5.1(d)); interior coordinates impose equalities and lower it.

on all 181 directions; in Example 5.5(i) no grid direction may be in Λ ; in (ii) the linear and exact tests must agree on all grid directions. (E3, concave) every cone point and every grid point in Λ_1 must be in Λ , and every grid point outside Λ_1 outside Λ . (E4, non-concave) the last statement must still hold (it is Theorem 5.1(b)); the number of certified witnesses of $\Lambda \subsetneq \Lambda_1$ is recorded, with no prediction. For Proposition 5.3, u_i' of the interior Cournot game is evaluated on a 401×401 grid of K and 61 weight directions; the maximum of W'_λ must not exceed $W'_\lambda(q^*) = 0$.

Results. (E2) Example 5.4: $x^* = (1, 1)$ with residual 0; agreement on 181/181 directions, 73 of which lie in the cone; for $\lambda = (1, 3)$ (outside the wedge) the exact maximiser is $x = (0.5, 1)$, better by 0.1250. Cournot (i): $q^* = (0.3333, 0.3333)$, 0/61 grid directions in Λ ; (ii): $q^* = (1, 1)$, cone of dimension 1, agreement 61/61, with 1 grid direction(s) in Λ_1 and 1 in Λ . Nikaidô–Isoda modification: maximum of W'_λ over the grid $\leq 2.8 \times 10^{-17}$ for all directions, so q^* is optimal for every λ (criterion met: yes). (E3–E4) Table 2 and Figure 3. In the 200 concave games, 61 (30%) have a non-trivial cone; all 268/268 cone points and all 882/882 in-cone grid points are in Λ , and all 24718/24718 out-of-cone grid points are outside Λ : equality $\Lambda = \Lambda_1$ as Theorem 5.1(c) predicts. In the 200 non-concave games, 103 have a non-trivial cone; the inclusion $\Lambda \subseteq \Lambda_1$ is never violated (23200/23200), but only 314/452 cone points and 2290/2400 in-cone grid points belong to Λ , and 58 instances (56% of those with a non-trivial cone) carry a certified witness of strict inclusion (for instance, in game N3 nonconcave with $x^* = (1.000, 1.000, 1.000)$ and $\lambda = (0.000, 0.552, 0.448) \in \Lambda_1$, the point $(0.000, 1.000, 1.000)$ improves W_λ by 2.15×10^{-1}). Maximum residuals 7.7×10^{-12} (concave) and 2.1×10^{-12} (non-concave); 51920 exact box maximisations in total.

7 What can be claimed

Table 3 lists every claim made in this note with its status. The reader in a hurry should retain three lines: the classical results are theorems and are proved above; the “formal relationship” between equilibrium and welfare is the first-order coincidence of Proposition 4.1, which is a definition rather than a mechanism; and (F, K) alone carries no welfare information (Proposition 5.3), so any welfare statement needs the externalities as separate data.

Claim	Status
Symmetric Jacobian on a convex open set gives a potential (Prop. 3.1). VI solutions of a gradient field are the stationary points of the potential, its minimisers if convex (Thm. 3.2).	classical; proved here classical (BMW 1956, Rosen 1965, Monderer–Shapley 1996); proved here
Nash and Wardrop equilibria are VI solutions (Lemmas 2.4, 2.7). Strict monotonicity: uniqueness; strong monotonicity and Lipschitz: existence and linear convergence of the projection iteration (Thm. 3.5).	classical; proved here classical (Kinderlehrer–Stampacchia, Facchinei–Pang); proved here
Pigou: PoA = 4/3 for $d = 1$, unbounded in d (Ex. 3.7).	classical; proved here; computed (E1a)
PoA $\leq 4/3$ for affine costs (Thm. 3.8).	classical (Roughgarden–Tardos 2002); proved here; verified on 200 instances
An equilibrium is a W -maximiser only if $\nabla W(x^*), -F(x^*) \in N_K(x^*)$; iff when W is concave; proportionality on a simplex face (Prop. 4.1). Marginal-cost tolls make the cost minimisers the equilibria; toll set $= -F(\bar{x}) - N_K(\bar{x})$ (Cor. 4.2).	proved here; elementary classical (Pigou, Knight, BMW, Hearn–Ramana); proved here; verified (E1c)
$\Lambda(x^*)$ is a closed convex cone, $\Lambda \subseteq \Lambda_1$, equality under joint concavity, block form, Pareto link (Thm. 5.1). (F, K) alone does not determine $\Lambda(x^*)$ (Prop. 5.3).	proved here; (a),(e) are standard scalarisation proved here; repackaging of Nikaidô–Isoda
Exact wedge cone in the two-player example (Ex. 5.4). $\Lambda = \Lambda_1$ in random concave games.	proved here; verified (E2) theorem; verified in 200/200 games (E3)
$\Lambda \subsetneq \Lambda_1$ occurs without joint concavity.	proved by certified witnesses in 58 games (E4); frequency is descriptive only
A new equilibrium concept or a general equilibrium solution. Open questions 8.1, 8.2.	not claimed open

Table 3: Claims and their status.

8 Limitations and open questions

Limitations. (1) Everything is finite-dimensional, deterministic and C^1 ; mixed strategies, atomic congestion games, dynamics and stochastic demand are outside the note. (2) Theorem 5.1 is a first-order statement; its only non-trivial sufficient condition for equality is concavity, which excludes Cournot-type payoffs. (3) The numerical experiments verify theorems on quadratic games over boxes and on parallel-link networks; the frequencies in Table 2 describe one sampling distribution and carry no general meaning. (4) Deciding $\lambda \in \Lambda(x^*)$ for a non-concave quadratic W_λ on a box is a non-convex quadratic programme, tractable here only because $N \leq 4$. (5) We have not surveyed the toll-pricing and multi-objective equilibrium literature exhaustively; the packaging of Theorem 5.1 may exist elsewhere.

Open question 8.1 (Second-order description of the cone). For quadratic payoffs on a polyhedral K , $\Lambda(x^*)$ is the zero set of the convex function $\lambda \mapsto \max_K W_\lambda - W_\lambda(x^*)$ on $\mathbb{R}_+^N$. Is there a finite description of $\Lambda(x^*)$ in terms of the data at x^* (a second-order refinement of Λ_1), and is $\Lambda(x^*)$ always polyhedral in this class? The witnesses of E4 show that Λ_1 is not that description.

Open question 8.2 (Commodity weights in routing). In a multi-commodity network let $C_k(f) = \sum_{p \in P_k} f_p c_p(f)$ be the cost borne by commodity k and f^* a Wardrop equilibrium. Theorem 5.1(a),(b) apply verbatim with players replaced by commodities ($u_k = -C_k$), but the C_k are not convex in general even for affine costs, so (c) does not. Characterise the cone of commodity weights $\lambda \in \mathbb{R}_+^m$ for which f^* minimises $\sum_k \lambda_k C_k$, and in particular decide whether it equals the first-order cone for affine separable costs.

Next steps. Answer Open question 8.1 at least for $N = 2$, where the face structure is small enough for a case analysis; run E4 with a global solver for larger N ; and confirm with the author whether the “formulation reviewed” in the line’s record was an instance of Proposition 4.1 or of Corollary 4.2, which is our reading.

A Reproducibility

The directory contains `experiments/vi_solver.py` (projections and the iteration of Theorem 3.5(b)), `experiments/traffic_poa.py` (E1) and `experiments/welfare_cone.py` (E2–E4). Seed 20260930; Python 3.11.15, NumPy 2.4.6 [8], SciPy 1.17.1 [22] (`linprog` with the HiGHS method for the cone geometry); E1 took 4 s and E2–E4 128 s on shared cores. The scripts write `results/results_traffic.json`, `results/results_welfare.json` and Markdown tables; `experiments/make_numbers.py` converts them into the macros and table bodies used here, so that every number in the text is generated.

Solver. Projection onto the box is coordinatewise clipping; projection onto the simplex uses the sort-based exact algorithm. The step is $\gamma = \mu/L^2$ with μ the least eigenvalue of the symmetric part of the Jacobian and L its spectral norm ($\mu = \min a_e$, $L = \max a_e$ for parallel links); the iteration stops when the step is below 10^{-12} (E1) or 10^{-13} (E2–E4).

Face enumeration. For each of the 3^N patterns in $\{0, 1, \text{free}\}^N$ the free coordinates solve $H_{ff}x_f = h_f - H_{fb}x_b$; candidates outside $[-10^{-9}, 1 + 10^{-9}]$ are discarded; the maximum over candidates is the maximum over the box.

Boundary detection. A coordinate within 10^{-9} of 0 or 1 is active (the projection returns exact bounds).

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